AgroVision AI

AI-Powered Plant Disease Detection & Smart Farmer Assistance

PBL / MTE Project Presentation

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TECHNOLOGY FOR
HEALTHIER CROPS
A BRIGHTER TOMORROW

PROBLEM STATEMENT

Plant diseases can significantly affect crop health and agricultural productivity, but identifying them early and accurately is difficult.

01

EARLY DETECTION

Disease symptoms can be subtle during the initial stages, making early identification difficult through visual inspection alone.

02

MANUAL DIAGNOSIS

Traditional identification depends heavily on human observation and agricultural expertise, which may not always be readily available.

03

ACTIONABLE GUIDANCE

Detecting a disease is only the first step; farmers also need understandable information about symptoms, treatment, prevention, and severity.

THE CORE CHALLENGE

Detecting disease is only the beginning — useful information must follow the prediction.

MOTIVATION

From disease classification to practical farmer assistance

01

Protect Crop Health

Early identification of visible disease symptoms can help farmers recognize potential crop-health problems sooner.

02

Make AI More Accessible

An image-based system can provide disease information without requiring the user to interpret technical symptoms or depend entirely on specialized expertise.

03

Turn Prediction into Action

A useful agricultural AI system should provide more than a disease label — it should present symptoms, treatment, prevention, and severity information in an understandable way.

OUR GOAL

To bridge the gap between AI-based disease recognition and practical, user-oriented agricultural assistance.

OBJECTIVES

To develop an AI-assisted plant disease detection and farmer-support system.

01

Disease Classification

Develop a deep-learning model capable of classifying plant leaf images into the supported disease and healthy classes.

02

Image-Based Prediction

Build an easy-to-use web interface where users can upload a leaf image and receive a disease prediction with its confidence score and Top-3 results.

03

Disease Understanding

Provide relevant information about the predicted condition, including symptoms, treatment, prevention, and severity.

04

Farmer Assistance

Generate practical recommendations for the user, including guidance in English and Hindi, based on the detected condition and severity.

05

Prediction Management

Maintain prediction history and provide analytics and reporting features to help users review previous results.

NOVELTY & KEY CONTRIBUTIONS

TRADITIONAL MODEL OUTPUT

Leaf Image → **Disease Class**
(only label)

AGROVISION AI

From prediction to practical farmer assistance

Validate
Check if image is a leaf

→

Predict
Disease class with confidence and Top-3 results

→

Explain
Symptoms, treatment, prevention, severity

→

Recommend
Practical guidance in English & Hindi

→

Record
History, analytics and reports

01 Integrated AI-to-Assistance Pipeline

Combines leaf validation, disease classification, confidence & Top-3 predictions, disease information and recommendations within one application.

02 Leaf Image Validation

A validation layer checks whether the uploaded image is sufficiently leaf-like before sending it to the classification model.

03 Beyond a Disease Label

Provides relevant information about the predicted condition, including symptoms, treatment, prevention, and severity category.

04 Contextual Farmer Recommendations

Generates practical recommendations based on the detected disease and severity, available in both English and Hindi.

05 Complete Application Ecosystem

Maintains prediction history and provides analytics, with options to export results as CSV and PDF reports.

The contribution lies in **integrating AI prediction with validation, disease information, contextual recommendations, and prediction management.**

PROPOSED SOLUTION

An AI-assisted pipeline for plant disease detection and farmer support.

01

Upload

User uploads a plant leaf image through the web application.

02

Validate

Leaf validation checks whether the image is suitable for analysis.

03

Predict

Deep-learning model classifies the image.

- Predicted class
- Confidence score
- Top-3 predictions

04

Understand

The system retrieves relevant disease information.

- Symptoms
- Treatment
- Prevention
- Severity category

05

Recommend

Recommendation Engine generates practical guidance based on the detected condition and severity.

- English guidance
- Hindi guidance

06

Record & Review

Prediction results are stored for future reference and analysis.

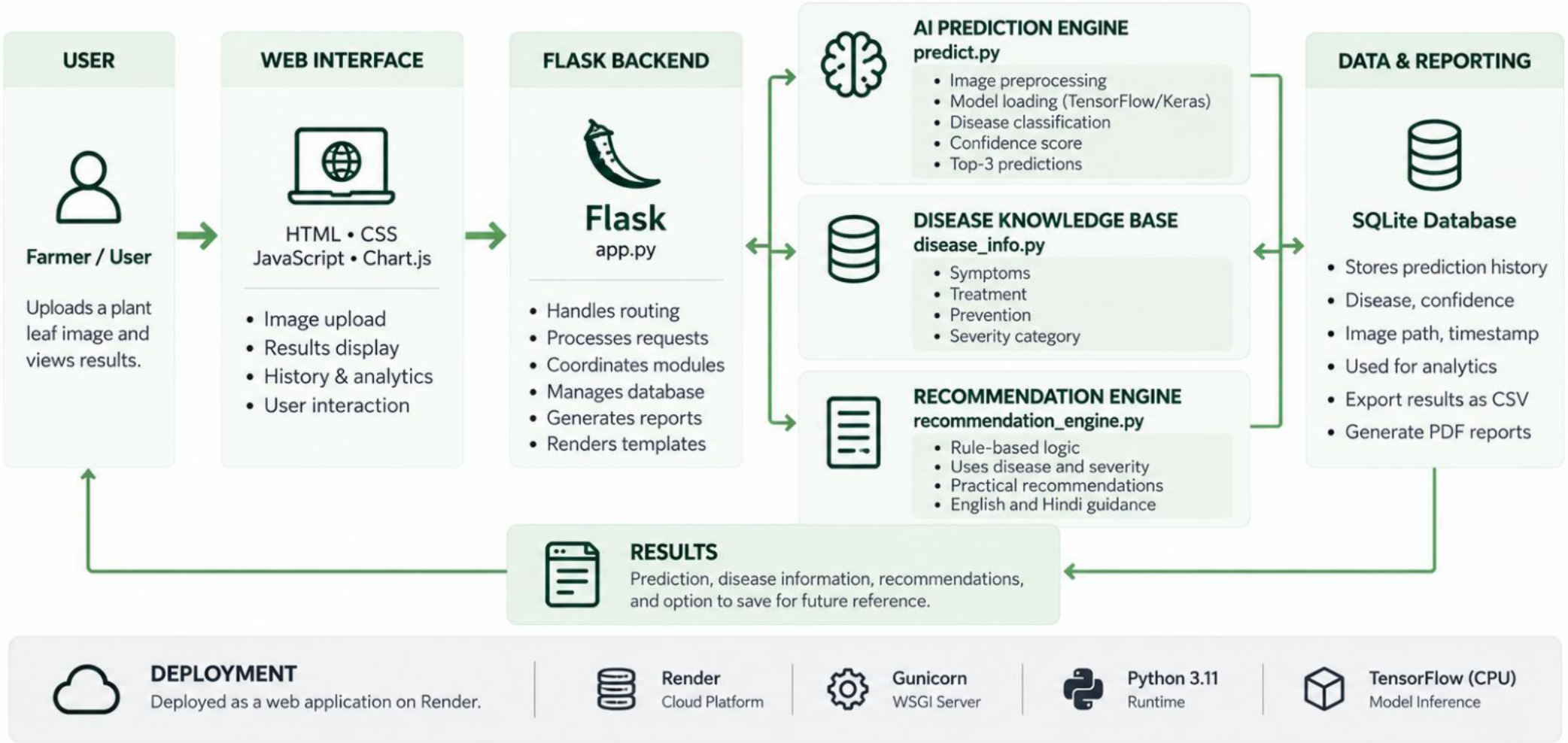
- History
- Analytics
- CSV export
- PDF reports

THE SOLUTION

AgroVision AI turns a leaf image into useful, actionable information — helping farmers identify plant diseases and take informed steps for crop care.

SYSTEM ARCHITECTURE

Modular architecture connecting the web interface, AI engine, knowledge base, recommendations, and data management.



DATASET — PLANTVILLAGE

A labeled image dataset used to train the plant-disease classification model.

PlantVillage Dataset

A widely used, labeled collection of plant leaf images suitable for supervised plant disease classification.

Why PlantVillage?

Selected because it provides labeled plant-leaf images covering multiple crops and disease conditions, making it suitable for supervised learning.

15 CLASSIFICATION CLASSES

The model classifies leaf images into 15 different classes across 3 crops.

Tomato
10 classes

1. Healthy
2. Bacterial Spot
3. Early Blight
4. Late Blight
5. Leaf Mold
6. Septoria Leaf Spot
7. Spider Mites
8. Target Spot
9. Tomato Mosaic Virus
10. Tomato Yellow Leaf Curl Virus

Potato
3 classes

1. Healthy
2. Early Blight
3. Late Blight

Bell Pepper
2 classes

1. Healthy
2. Bacterial Spot

TRAIN / VALIDATION SPLIT

The dataset is divided into training and validation sets in an 80/20 ratio.

80%	20%
Training Set (80% of data)	**Validation Set** (20% of data)

FINAL EVALUATION

Model performance is reported on the held-out validation set used for final evaluation.

4,134 images
Used for final evaluation and reported metrics.

"Tomato • Potato • Bell Pepper | 15 Classes | Real Leaf Images | For a Healthier Tomorrow"

DATA PREPROCESSING & AUGMENTATION

Preparing leaf images for robust deep-learning training.

01 IMAGE RESIZING

All input images are resized to 224 × 224 pixels.

Original image (variable size)

Resized image 224 × 224

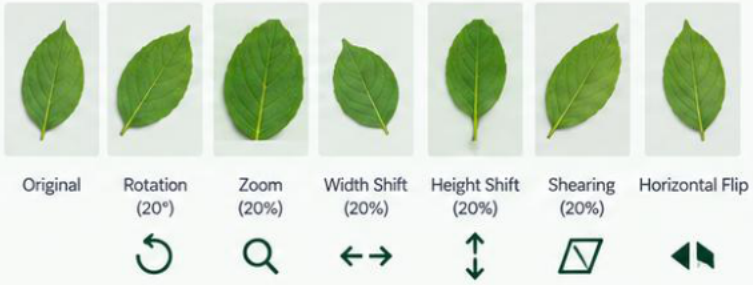
02 EFFICIENTNET PREPROCESSING

Images are prepared using EfficientNet's preprocess_input to convert them into the required numerical format for the pretrained network.

Preprocessed image (ready for augmentation)

03 DATA AUGMENTATION

Applied during training only to introduce controlled image variation.



Augmentation Techniques Used:
Rotation, Zoom, Width Shift, Height Shift, Shearing and Horizontal Flip (with typical range of 20%).

04 MODEL-READY INPUT

Preprocessed (and augmented for training) images are ready to be fed into EfficientNetB0.

224 × 224 × 3

EfficientNetB0

Training set: Preprocessing + random augmentation

Validation set: Preprocessing only (no augmentation)

Same leaves.
More possibilities.

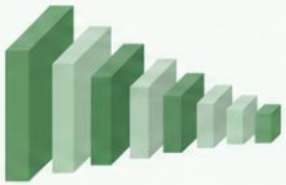
DEEP LEARNING MODEL — EFFICIENTNETB0

Transfer learning with a pretrained EfficientNetB0 backbone for plant disease classification.

INPUT IMAGE

224 × 224 × 3
(RGB Image)

EFFICIENTNETB0 (BACKBONE)



Pretrained on ImageNet
include_top = False

Extracts rich visual features from the leaf image.

GLOBAL AVERAGE POOLING

GlobalAveragePooling2D

Converts spatial feature maps into a compact feature representation.

DENSE LAYER

Dense (256)
ReLU

Learns higher-level feature combinations for classification.

DROPOUT

Dropout (0.4)

Learns high disables a fraction of neurons during training to improve generalization.

OUTPUT LAYER

Dropout (15)
Softmax

Outputs probability scores for 15 classes (plant diseases and healthy leaves).

PREDICTION

Tomato - Early Blight	0.78
Tomato - Healthy	0.12
Tomato - Late Blight	0.06

(Top-3 predictions)

WHY EFFICIENTNETB0?

- Provides strong feature extraction with fewer parameters than larger models.
- Pretrained on ImageNet, allowing effective transfer learning.
- Well-suited for image classification tasks and relatively lightweight.
- Helps achieve good performance even with limited domain-specific data.

WHAT IS TRANSFER LEARNING?

Instead of learning all visual features from scratch, we use a model pretrained on the large ImageNet dataset and adapt it to our plant disease classification task by adding a custom classification head and fine-tuning on PlantVillage images.

MODEL CONFIGURATION (SUMMARY)

Backbone	Pretraining	Input Size	Top Classifier	Pooling	Dense Layer	Dropout	Output Layer
EfficientNetB0	ImageNet	224 × 224 × 3	Removed (include_top = False)	Global Average Pooling	256 (ReLU)	0.4	15-class Softmax

KEY TAKEAWAY | EfficientNetB0 extracts visual features from the leaf image, while the custom classification head maps those features to 15 plant-health classes.

TRAINING & FINE-TUNING

Adapting the pretrained EfficientNetB0 model to plant disease classification.

1 TRAINING DATA PIPELINE

Training images are preprocessed and augmented before being fed to the model.

PlantVillage training images

Resize
224 × 224

EfficientNet preprocessing

Random augmentation

2 TWO-STAGE TRAINING STRATEGY

Initial training followed by fine-tuning.

STAGE 1 Initial Training

Train the classification head on top of the pretrained EfficientNetB0 features.

- Use pretrained EfficientNetB0 (ImageNet weights)
- Classification head learns plant disease patterns
- Base layers kept frozen during this stage

STAGE 2 Fine-Tuning

Unfreeze selected deeper layers and fine-tune the model with a smaller learning rate.

- Fine-tune deeper layers of EfficientNetB0
- Learning rate: 1×10^{-5}
- Adapt pretrained features to the plant disease domain

3 VALIDATION PIPELINE

Validation images are only preprocessed (without augmentation).

Validation images

Resize
224 × 224

EfficientNet preprocessing

4 TRAINED MODEL

Outputs probability scores for 15 classes.

Trained EfficientNetB0 Model

15-class Softmax

Predicted Class
(Disease / Healthy)

TRAINING CONFIGURATION

Key parameters used for training the model.

Optimizer	Adam
Loss Function	Categorical Cross-Entropy
Batch Size	32
Input Size	224 × 224 × 3
Output	15-class Softmax

TRAINING CONTROL

Callbacks used to monitor and save the best model.

Early Stopping

- Monitors validation loss
- Stops training when further improvement is no longer observed

Model Checkpoint

- Saves the best-performing model during training
- Ensures the best weights are retained for evaluation and deployment

WHY A SMALL LEARNING RATE FOR FINE-TUNING?

A lower learning rate (1×10^{-5}) allows gradual updates to the pretrained features, helping retain useful ImageNet knowledge while adapting the model to plant disease classification.

KEY TAKEAWAY | We use a two-stage training approach with transfer learning, Adam optimizer, and a small learning rate for fine-tuning to build a robust plant disease classifier.

EXPERIMENTAL RESULTS

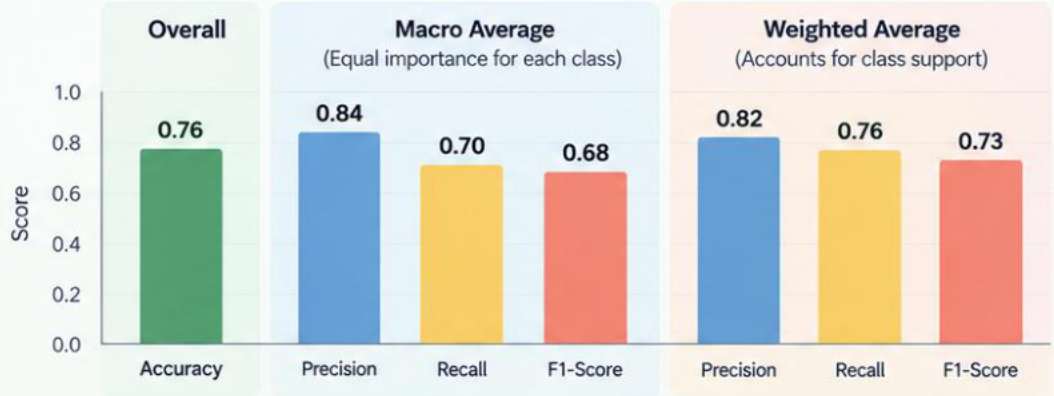
Evaluation of the trained EfficientNetB0 classifier on 4,134 images.

OVERALL ACCURACY

Evaluated on
4,134 images
across 15 classes

EVALUATION METRICS

Precision, Recall and F1-score for macro and weighted averages.



EVALUATION SET DETAILS

Dataset	PlantVillage
Total images	4,134
Number of classes	15
Evaluation	Classification report
Metrics	Accuracy, Precision, Recall, F1-score

Detailed class-wise performance is shown in the next slide.

MACRO vs WEIGHTED AVERAGE

Macro Average

- Calculates the metric for each class and gives every class equal importance.
- Useful for understanding overall model performance across all classes.

Weighted Average

- Calculates the metric across classes while accounting for the number of samples in each class.
- More influenced by classes with higher number of images.

KEY OBSERVATIONS

- The model achieves 76% overall accuracy on 4,134 images.
- Macro precision (84%) is relatively high, but macro recall (70%) and macro F1 (68%) indicate that some classes are more challenging to detect.
- Performance varies considerably between classes due to differences in visual similarity and class support.

WHAT THE METRICS TELL US

Precision (84% macro)
When the model predicts a class, it is correct most of the time (on average across classes).

Recall (70% macro)
The model is able to detect a substantial portion of actual instances, but misses some, especially in more challenging classes.

F1-Score (68% macro)
Reflects the balance between precision and recall. The lower macro F1 shows the difficulty of maintaining strong performance across all 15 classes.

CLASS-WISE PERFORMANCE ANALYSIS

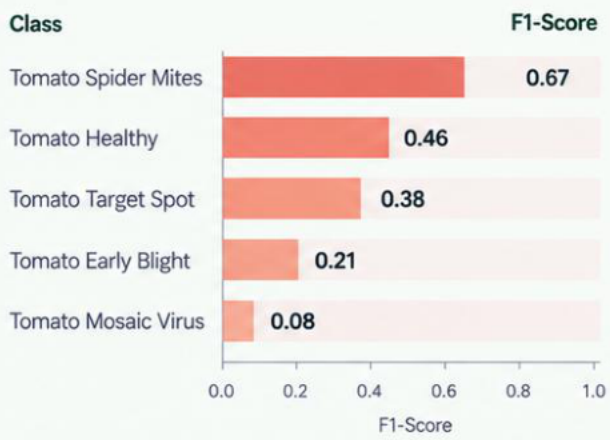
Performance varies across diseases despite the 76% overall accuracy.

BEST PERFORMING CLASSES

Selected classes with high F1-scores.

LOWER PERFORMING CLASSES

Selected classes with lower F1-scores.



BEST PERFORMING EXAMPLE

Tomato Yellow Leaf Curl Virus

Precision	0.97
Recall	0.96
F1-Score	0.97
Support	642

The model accurately identifies this disease in most cases.

MOST CHALLENGING EXAMPLE

Tomato Mosaic Virus

Precision	1.00
Recall	0.04
F1-Score	0.08
Support	75

High precision but very low recall — only a small fraction of actual cases are detected.

WHY F1-SCORE?

F1-score combines Precision and Recall into a single measure, making it useful for comparing performance across classes, especially when performance is uneven.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

MACRO vs WEIGHTED AVERAGE

Macro Average

- Calculates the metric for each class and gives every class equal importance.
- Useful for understanding per-class performance.

Weighted Average

- Calculates the metric across classes while accounting for the number of samples in each class.
- More influenced by classes with higher support.

KEY OBSERVATION

- Performance is highly class-dependent.
- Some classes achieve high F1-scores, while others are substantially lower.
- The 76% overall accuracy masks significant variation between individual diseases.

POSSIBLE FACTORS FOR CHALLENGING CLASSES

- Visual similarity between different diseases.
- Variation in leaf appearance, size and background.
- Differences in class distribution (number of samples).
- Challenges in learning distinct features for visually similar conditions.

"Different leaves, different challenges — towards more accurate and reliable plant disease detection."

AgroVision AI | MTE Project

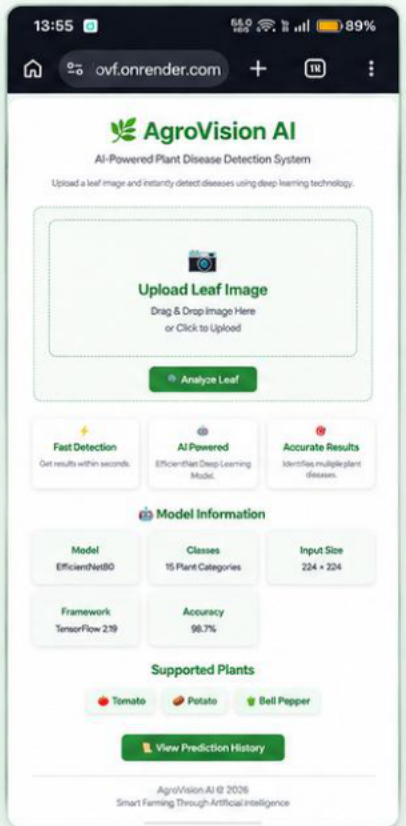
WEB APPLICATION — AGROVISION AI

Turning the trained model into an accessible web-based plant disease detection system.

"From Leaf Image to Farmer Guidance"

01 UPLOAD LEAF IMAGE

User uploads a plant leaf image for analysis.



- Simple and clean interface
- Quick analysis
- Supports Tomato, Potato and Bell Pepper

02 AI PREDICTION RESULT

Displays the detected disease, confidence, severity, Top-3 predictions and detailed information.

- Detected disease name
- Confidence and reliability
- Disease severity level
- Top-3 predictions
- Symptoms, treatment and prevention
- Bilingual recommendations (English + Hindi)
- Download PDF report

03 USER HISTORY & ANALYTICS

Track past analyses and view overall statistics.

Prediction History

View past predictions with date, time and confidence. Export history as CSV.

Analytics Dashboard

Visualize prediction statistics including healthy vs diseased plants and average confidence.

END-TO-END PIPELINE

Upload Image → Preprocess & Validate → EfficientNetB0 Model → Prediction Result → Farmer Recommendation

TECHNOLOGY STACK

Flask | TensorFlow/Keras | HTML/CSS/JavaScript | Chart.js

KEY APPLICATION FEATURES

From disease prediction to complete analysis, recommendation, and prediction management.

"A complete solution for healthier crops"

01 LEAF VALIDATION

Checks if the uploaded image is suitable for plant-leaf analysis.

- Detects whether the image contains a plant leaf
- Helps prevent invalid or unrelated images
- Ensures better prediction reliability

Valid leaf image → proceed to analysis

02 CONFIDENCE & TOP-3 PREDICTIONS

Provides the model's prediction confidence and the top 3 possible classes.

- Predicted disease with confidence score
- Reliability indicator (e.g. Low/Medium/High)
- Top-3 predictions with probabilities
- Helps users understand prediction certainty

More transparent and reliable results

03 DISEASE INFORMATION

Displays detailed information about the detected disease.

- Disease name and description
- Common symptoms
- Recommended treatment
- Preventive measures
- Severity information (Low / Medium / High)

Helps users learn about plant diseases

04 FARMER RECOMMENDATIONS

Rule-based recommendation engine provides practical guidance.

- Personalized recommendations based on disease and severity
- Guidance in English
- Guidance in Hindi
- Easy-to-follow and farmer-friendly suggestions

Actionable advice for healthier crops

05 PREDICTION HISTORY

Automatically saves all past analyses for future reference.

- Stores predicted disease
- Saves confidence score
- Records image and timestamp
- Built using SQLite database
- Allows users to revisit previous predictions

Never lose your analysis history

06 ANALYTICS DASHBOARD

Visualizes overall prediction statistics and insights.

- Total number of analyses
- Healthy vs diseased plants count
- Average confidence
- Disease distribution (pie chart)
- Helps track usage and trends over time

Understand your crop health trends

07 PDF REPORT

Generates a downloadable analysis report for each prediction.

- Includes disease name and image
- Confidence score and severity
- Symptoms, treatment and prevention
- Timestamp of analysis
- Generated using ReportLab

Download and share detailed reports

08 CSV EXPORT

Export prediction history for further analysis.

- Download entire prediction history as a CSV file
- Includes disease, confidence, image path and timestamp
- Useful for personal records and external analysis
- Generated dynamically

Easily export your data

AGROVISION AI
Smart Farming Through Artificial Intelligence

"AI Insights. Practical Solutions. Healthier Harvests."

Built for Farmers.
Powered by AI.

LEAF VALIDATION & INPUT SAFETY

AgroVision AI
Smart Farming Through Artificial Intelligence

HEALTHY PLANTS
BRIGHTER FUTURES

Preventing unsuitable images from being passed directly to the disease classifier.

VALIDATION PIPELINE

A lightweight image-validation layer based on image characteristics (OpenCV).

1. UPLOAD IMAGE

User uploads an image from device.

2. READABILITY & SIZE CHECK

Check if image is readable and valid format.

3. RESIZE TO 224 × 224

Image is resized to 224 × 224 pixels (for analysis).

4. HSV / IMAGE ANALYSIS

Analyze image characteristics using HSV color space and edges.

5. LEAF VALIDATION SCORE (0 – 5)

Combine all indicators to get a validation score (0 – 5).

6. SCORE ≥ 3/5 ?

YES

PROCEED TO CNN

EFFICIENTNETB0
(15 classes)

Disease prediction with confidence and Top-3 results.

NO

REJECT IMAGE

Image is not a valid plant leaf. Do not pass to classifier.

IMAGE CHARACTERISTICS USED FOR VALIDATION

The validation score is based on simple but effective image characteristics.

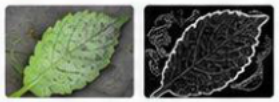
GREEN PIXEL RATIO

- Convert image to HSV and detect green regions.
- Green ratio ≥ 0.15 → +2
- Green ratio ≥ 0.30 → +1 (additional)

SATURATION RATIO

- Measure the proportion of sufficiently saturated pixels.
- Saturation ratio ≥ 0.20 → +1

EDGE / TEXTURE RATIO



- Convert to grayscale and apply Canny edge detection.
- Edge ratio ≥ 0.03 → +1

SCORING & DECISION

Scoring Criteria	
Green ratio ≥ 0.15	+2 points
Green ratio ≥ 0.30	+1 point (additional)
Saturation ratio ≥ 0.20	+1 point
Edge ratio ≥ 0.03	+1 point

Total score: 0 – 5

If score < 3 → Reject image

If score ≥ 3 → Proceed to CNN

WHY VALIDATE?

- Our model is a closed-set classifier with 15 classes, so an arbitrary image can still be forced into one of the disease classes.
- Validation acts as a lightweight input gate before CNN inference.
- Helps prevent invalid or unrelated images, leading to more reliable predictions.

REJECTION MESSAGE (SHOWN TO USER)

"This does not appear to be a plant leaf.
Please upload a clear image of a tomato, potato, or bell pepper leaf."

LIMITATIONS

Current boundaries of the AgroVision AI system.

CURRENT LIMITATIONS

Important boundaries of the current implementation

01 **PlantVillage-Based Training**
The model is trained on PlantVillage images; real-world field conditions may differ from the dataset.

02 **Limited Crop Coverage**
The current classifier supports Tomato, Potato and Bell Pepper across 15 classes.

03 **Classification Only**
The system predicts a disease class but does not localize the exact affected region of the leaf.

04 **No Infection Percentage**
The system does not calculate the percentage of leaf area affected by disease.

05 **No Explainable AI Visualization**
Grad-CAM or similar visual explanations are not currently implemented.

CONCLUSION

AgroVision AI
Smart Farming Through Artificial Intelligence

HEALTHY PLANTS
HEALTHY FARMERS
BRIGHTER TOMORROWS

An end-to-end AI-assisted platform for plant disease detection and farmer support.

AgroVision AI integrates deep-learning-based plant disease classification with image validation, disease information, farmer recommendations, and prediction management into a single web-based platform.

WHAT WE DEVELOPED

AI Disease Classification
EfficientNetB0 with transfer learning for 15 plant disease/healthy classes.

Input Validation
Lightweight leaf-image validation before classification.

Farmer Assistance
Disease symptoms, treatment, prevention, severity information and rule-based recommendations.

Web Application
Upload, prediction, Top-3 results, history, analytics, CSV export and PDF reporting.

VERIFIED RESULTS

76%
Evaluation Accuracy

4,134 evaluation images • 15 classes

Macro F1: 68% • Weighted F1: 73%

FROM MODEL → APPLICATION

EfficientNetB0
Trained model with transfer learning

→ **Flask Web App**
User-friendly web interface

→ **Leaf Validation**
Input safety check

→ **Prediction**
Disease class with confidence

→ **Disease Information + Recommendations**
Symptoms, treatment, prevention, severity

→ **History / Analytics / Reports**
Save, visualize and export results

Deployment: Render • Gunicorn • Python 3.11 • TensorFlow/Keras

AgroVision AI demonstrates how a plant-disease classification model can be integrated into a practical, user-oriented agricultural assistance system.

Different Leaves
Healthier Tomorrows

FUTURE WORK

Potential extensions to improve generalization, explainability, and deployment.

FUTURE SCOPE

Planned directions for extending the current system

01 Larger & Real-World Datasets
Add diverse field images and larger datasets to improve model generalization beyond curated images.

02 Expand Crop Coverage
Support additional crops and a wider range of plant diseases.

03 Grad-CAM Integration
Add visual explanations to highlight image regions that influence the model's prediction.

04 Disease Localization & Severity Estimation
Use segmentation or localization to identify affected regions and estimate severity from lesion area.

05 Mobile / Edge AI Deployment
Explore TensorFlow Lite and model quantization for efficient on-device inference.

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